

NETWORKS AND COMPLEXITY

Solution 22-2

*This is an example solution from the forthcoming book Networks and Complexity.
Find more exercises at <https://github.com/NC-Book/NCB>*

Ex 22.2: Kronecker rules [Lvl. 1]

Show either generally or with the matrices from the previous exercise that

$$\text{a) } c(\mathbf{A} \otimes \mathbf{C}) = c\mathbf{A} \otimes \mathbf{C} = \mathbf{A} \otimes c\mathbf{C}, \text{ b) } (\mathbf{A} \otimes \mathbf{C})(\mathbf{u} \otimes \mathbf{w}) = \mathbf{A}\mathbf{u} \otimes \mathbf{C}\mathbf{w},$$

where c is a scalar and $\mathbf{u}$ and $\mathbf{w}$ are vectors of suitable dimension.

Solution

Explicit Matrices. Let's first do this with the matrices from the previous exercise. We start with property (a)

$$c(\mathbf{A} \otimes \mathbf{C}) = c \left[\begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix} \otimes \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} \right] \quad (1)$$

$$= c \begin{pmatrix} 1 & 0 & 2 & 0 \\ 0 & 0 & 0 & 0 \\ 3 & 0 & 4 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix} \quad (2)$$

$$= \begin{pmatrix} 1c & 0 & 2c & 0 \\ 0 & 0 & 0 & 0 \\ 3c & 0 & 4c & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix} \quad (3)$$

$$= \left[\begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix} \otimes \begin{pmatrix} c & 0 \\ 0 & 0 \end{pmatrix} \right] = \mathbf{A} \otimes c\mathbf{C} \quad (4)$$

Reusing the first three steps we can also write

$$c(\mathbf{A} \otimes \mathbf{C}) = \begin{pmatrix} 1c & 0 & 2c & 0 \\ 0 & 0 & 0 & 0 \\ 3c & 0 & 4c & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix} = \left[\begin{pmatrix} 1c & 2c \\ 3c & 4c \end{pmatrix} \otimes \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} \right] = c\mathbf{A} \otimes \mathbf{C} \quad (5)$$

To show the second property (b) we first compute

$$(\mathbf{A} \otimes \mathbf{C})(\mathbf{u} \otimes \mathbf{w}) = \left[\begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix} \otimes \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} \right] \left[\begin{pmatrix} u_1 \\ u_2 \end{pmatrix} \otimes \begin{pmatrix} w_1 \\ w_2 \end{pmatrix} \right] \quad (6)$$

$$= \begin{pmatrix} 1 & 0 & 2 & 0 \\ 0 & 0 & 0 & 0 \\ 3 & 0 & 4 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix} \begin{pmatrix} u_1 w_1 \\ u_1 w_2 \\ u_2 w_1 \\ u_2 w_2 \end{pmatrix} \quad (7)$$

$$= \begin{pmatrix} (u_1 + 2u_2)w_1 \\ 0 \\ (3u_1 + 4u_2)w_1 \\ 0 \end{pmatrix}. \quad (8)$$

Instead of trying to go from here back to a matrix product, let's consider the right side of the equation we are trying to prove:

$$\mathbf{A}\mathbf{u} \otimes \mathbf{C}\mathbf{w} = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix} \begin{pmatrix} u_1 \\ u_2 \end{pmatrix} \otimes \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} \begin{pmatrix} w_1 \\ w_2 \end{pmatrix} \quad (9)$$

$$= \begin{pmatrix} u_1 + 2u_2 \\ 3u_1 + 4u_2 \end{pmatrix} \otimes \begin{pmatrix} w_1 \\ 0 \end{pmatrix} \quad (10)$$

$$= \begin{pmatrix} (u_1 + 2u_2)w_1 \\ 0 \\ (3u_1 + 4u_2)w_1 \\ 0 \end{pmatrix}, \quad (11)$$

which is the same result we found previously, so the two expressions are equal.

General case. To show the relationships in the general case let's think of the result of the Kronecker product $\mathbf{R} = \mathbf{A} \otimes \mathbf{C}$ as a block matrix, we can then define

$$R_{ni,mj} = [\mathbf{A} \otimes \mathbf{C}]_{ni,mj} \quad (12)$$

as the element i, j in the block n, m . In this notation the Kronecker product is defined as

$$[\mathbf{A} \otimes \mathbf{C}]_{ni,mj} = A_{n,m} C_{i,j} \quad (13)$$

To show property (a) consider the element ni, mj of $c(\mathbf{A} \otimes \mathbf{C})$, which we write as

$$[c(\mathbf{A} \otimes \mathbf{C})]_{ni,mj} = c[(\mathbf{A} \otimes \mathbf{C})]_{ni,mj} = cA_{n,m} C_{i,j} \quad (14)$$

Likewise we can write

$$[c\mathbf{A} \otimes \mathbf{C}]_{ni,mj} = [c\mathbf{A} \otimes \mathbf{C}]_{ni,mj} = [c\mathbf{A}]_{n,m} C_{i,j} = cA_{n,m} C_{i,j} \quad (15)$$

and also

$$[\mathbf{A} \otimes c\mathbf{C}]_{ni,mj} = [\mathbf{A} \otimes c\mathbf{C}]_{ni,mj} = A_{n,m} [c\mathbf{C}]_{i,j} = cA_{n,m} C_{i,j}. \quad (16)$$

The three results are the same. So we have shown

$$[c(\mathbf{A} \otimes \mathbf{C})]_{ni,mj} = [c\mathbf{A} \otimes \mathbf{C}]_{ni,mj} = [\mathbf{A} \otimes c\mathbf{C}]_{ni,mj} \quad (17)$$

Because this is true independently of the values of the indices n, m, i , and j , it must also be true for the entire matrix, which proves

$$c(\mathbf{A} \otimes \mathbf{C}) = c\mathbf{A} \otimes \mathbf{C} = \mathbf{A} \otimes c\mathbf{C}. \quad (18)$$

To show the second property (b) we need a blockwise version of the inner product between matrices. For non block matrices the inner product is defined by the element-wise rule

$$[\mathbf{A}\mathbf{C}]_{i,j} = \sum_k A_{i,k} C_{k,j}. \quad (19)$$

Similarly we can multiply blockwise matrices by multiplying the respective blocks

$$[\mathbf{A}\mathbf{C}]_{m,n} = \sum_l \mathbf{A}_{m,l} \mathbf{C}_{l,n}. \quad (20)$$

which means

$$[\mathbf{AC}]_{mi,nj} = \left[\sum_l \mathbf{A}_{m,l} \mathbf{C}_{l,n} \right]_{i,j} . \quad (21)$$

$$= \sum_l [\mathbf{A}_{m,l} \mathbf{C}_{l,n}]_{i,j} \quad (22)$$

$$= \sum_{l,k} A_{mi,lk} C_{lk,nj} \quad (23)$$

This is the same as the simple element-wise formula above, only the summation index is now the combined index $lk = l, k$ instead of just k .

We can now write the left side of the relationship that we want to prove as

$$[(\mathbf{A} \otimes \mathbf{C})(\mathbf{u} \otimes \mathbf{w})]_{mi,nj} = \sum_{l,k} [\mathbf{A} \otimes \mathbf{C}]_{mi,lk} [\mathbf{u} \otimes \mathbf{w}]_{lk,nj} \quad (24)$$

$$= \sum_{l,k} A_{mi,l} C_{i,k} u_{l,n} w_{k,j} \quad (25)$$

where we used the definition of the Kronecker product in the second step. Now consider

$$[(\mathbf{A}\mathbf{u} \otimes \mathbf{C}\mathbf{w})]_{ni,mj} = [\mathbf{A}\mathbf{u}]_{m,n} [\mathbf{C}\mathbf{w}]_{i,j} \quad (26)$$

$$= \left(\sum_l A_{m,l} u_{l,n} \right) \left(\sum_k C_{i,k} w_{k,j} \right) \quad (27)$$

$$= \sum_{l,k} A_{m,l} C_{i,k} u_{l,n} w_{k,j}. \quad (28)$$

Again the two expressions are identical so

$$[(\mathbf{A} \otimes \mathbf{C})(\mathbf{u} \otimes \mathbf{w})]_{mi,nj} = [(\mathbf{A}\mathbf{u} \otimes \mathbf{C}\mathbf{w})]_{ni,mj}. \quad (29)$$

Since this is true for every element, it must also be true for the entire matrices, which shows

$$(\mathbf{A} \otimes \mathbf{C})(\mathbf{u} \otimes \mathbf{w}) = (\mathbf{A}\mathbf{u} \otimes \mathbf{C}\mathbf{w}). \quad (30)$$

Note that this proof also applies to the case where $\mathbf{u}$ and $\mathbf{w}$ are matrices.